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SECTOR AT A GLANCE

Farm Based Livelihoods

Skilling, Livelihoods and Labour Welfare Sector in India

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Labour Welfare Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Farm based Livelihoods. NITI Aayog <Sector-at-a-Glance> Series – Skilling, Livelihoods, and Labour Welfare Sector in India, 2023-Skilling, Livelihoods and Labour Welfare-SAG-05, <link>

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Overview

The farm-based livelihoods sub-sector plays a significant role in the socio-economic fabric of India. The agriculture and allied sectors which largely includes agriculture, animal husbandry and aquaculture, form the bedrock of the farm-based livelihoods and the Indian economy. It engages more than 50% of the Indian workforce and contributes **21.1% of the Gross Value Added (GVA)** to the economy at Rs 22.2 lakh crores with a **growth rate of 3.3%.**¹

The sector contributes a share of **11.9%** to **the total national exports at Rs 3.7 lakh crores.**² It not only provides livelihoods opportunities to more than half of the Indian population but also ensures food security to the nation, rural development and poverty alleviation.

Figure 1: Gross Value Added (GVA) to the Indian Economy*



Source: Agriculture statistics at a Glance 2022 (*at constant (2011-12) basic prices)

*GVA by Agriculture and aquaculture sector encompasses the GVA by food processing sector

Since the agriculture and allied sector contributes significantly to the farm-based livelihoods ecosystem, this document encompasses its broad sub-sectors viz. agriculture, animal husbandry and aquaculture. The farm-based livelihoods landscape of India is marked by small and marginal farmers (with a total operational land holding less than 2 hectares), which forms **86.2%** of the total farmers in the country.³ The Indian agriculture sector with its diverse agro-climatic zones sustains wide array of farm-level production ranging from staple grains to cash crops and all major kinds of animal husbandry and aquaculture practices. India has the **world's largest herd of bovines** (cattle and buffaloes), largest area planted to wheat, rice, and cotton, largest production of inland-capture fisheries and is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruit, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar.⁴

¹ Department of Agriculture & Farmers Welfare, MoA & FW, Government of India, India (2022). [Agricultural Statistics at a Glance 2022](#)

² Department of Agriculture & Farmers Welfare, MoA & FW, Government of India, India (2022). [Agricultural Statistics at a Glance 2022](#)

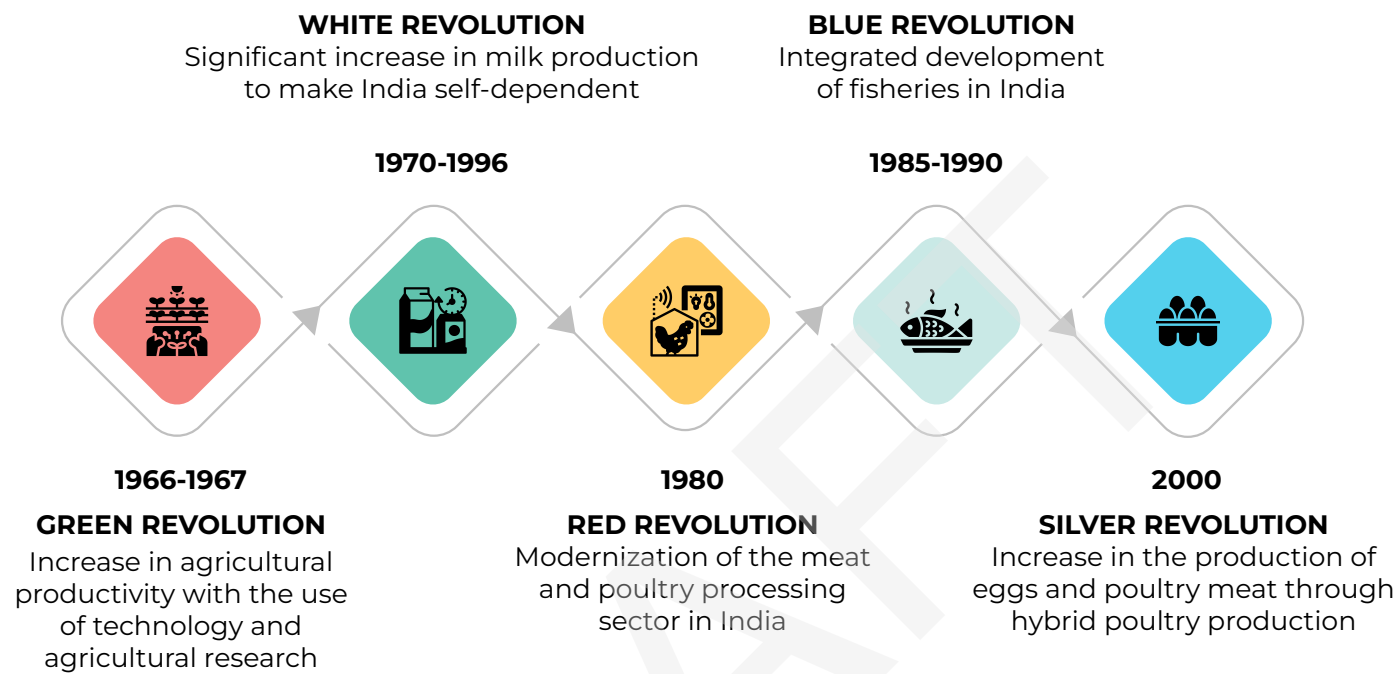
³ NABARD (2022). [Aggregation model of sustainable farming for smallholder farmers for the Asia Specific Region](#)

⁴ [Agriculture in India : FAQ in India \(n.d.\). India at a Glance](#)

Sub-Sector Evolution

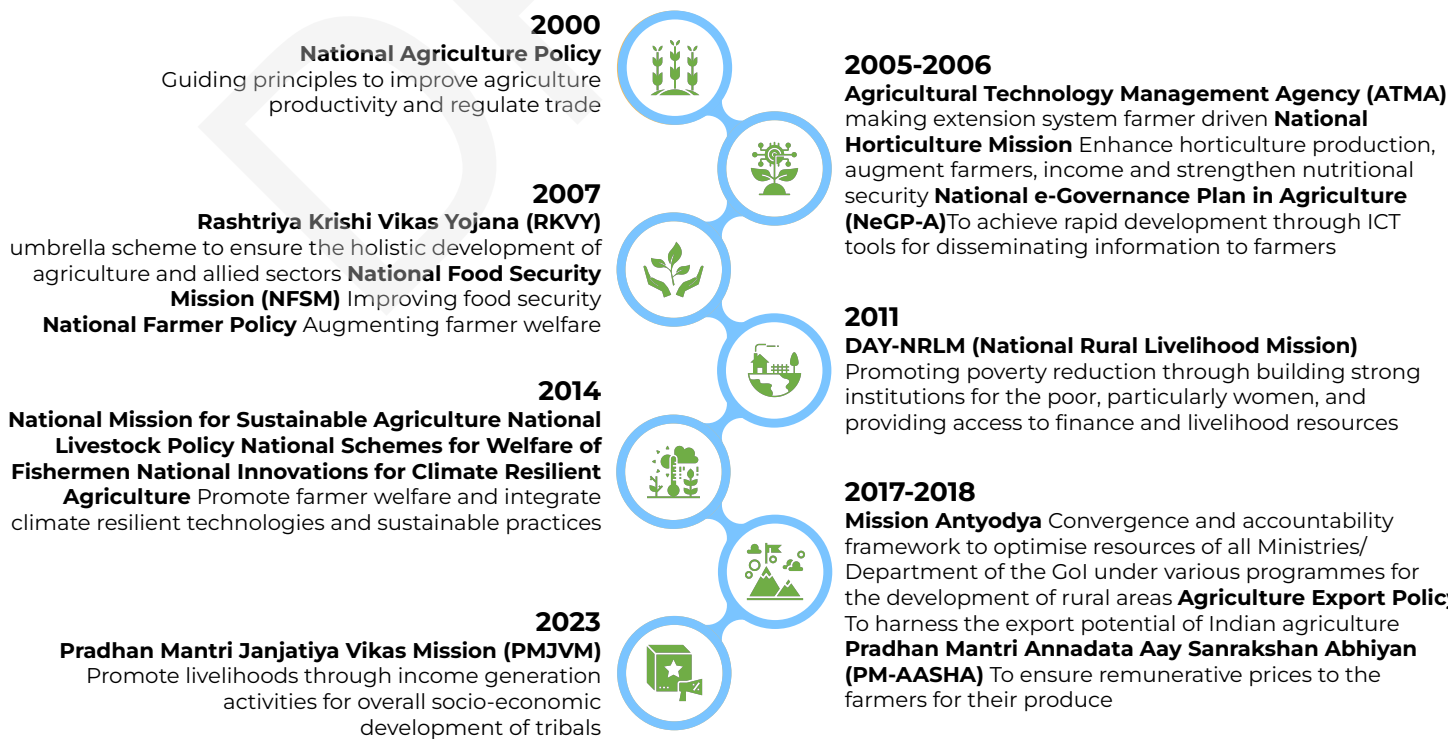
Pre-Independence, the nation faced challenges related to nutritious food security as traditional or natural farming methods were in practice. Post-Independence, following major revolutions in the agriculture and allied sector have transformed nationwide farm productivity, income and livelihoods.⁵

Figure 2: Major revolutions in the agriculture and allied sector



Over the last two decades, the following **major policy level initiatives** have bolstered the farm-based livelihoods ecosystem in the country:

Figure 3: Major policy level initiatives over the years



⁵FAO (2004).The state of food insecurity in the world (SOFI)

Stakeholder's Mapping

Following figure highlights the **key stakeholders** of farm-based livelihoods ecosystem:

Figure 4: Key stakeholders of farm-based livelihoods ecosystem

CENTRAL MINISTRIES

- Ministry of Rural Development (Nodal Ministry)
- Ministry of Fisheries, Animal Husbandry and Dairying
- Ministry of Food Processing
- Ministry of Agriculture & Farmers' Welfare
- Other Ministries

ENABLERS

- National Rural Livelihood Mission (NRLM)
- National Livestock Mission (NLM)
- National Institute of Rural Development & Panchayati Raj (NIRD & PR)
- National Dairy Development Board (NDDB)
- SPM Rurban Mission
- National Bank for Agriculture and Rural Development (NABARD)
- Civil Society organizations
- Financial Institutions and other private entities



IMPLEMENTING BODIES

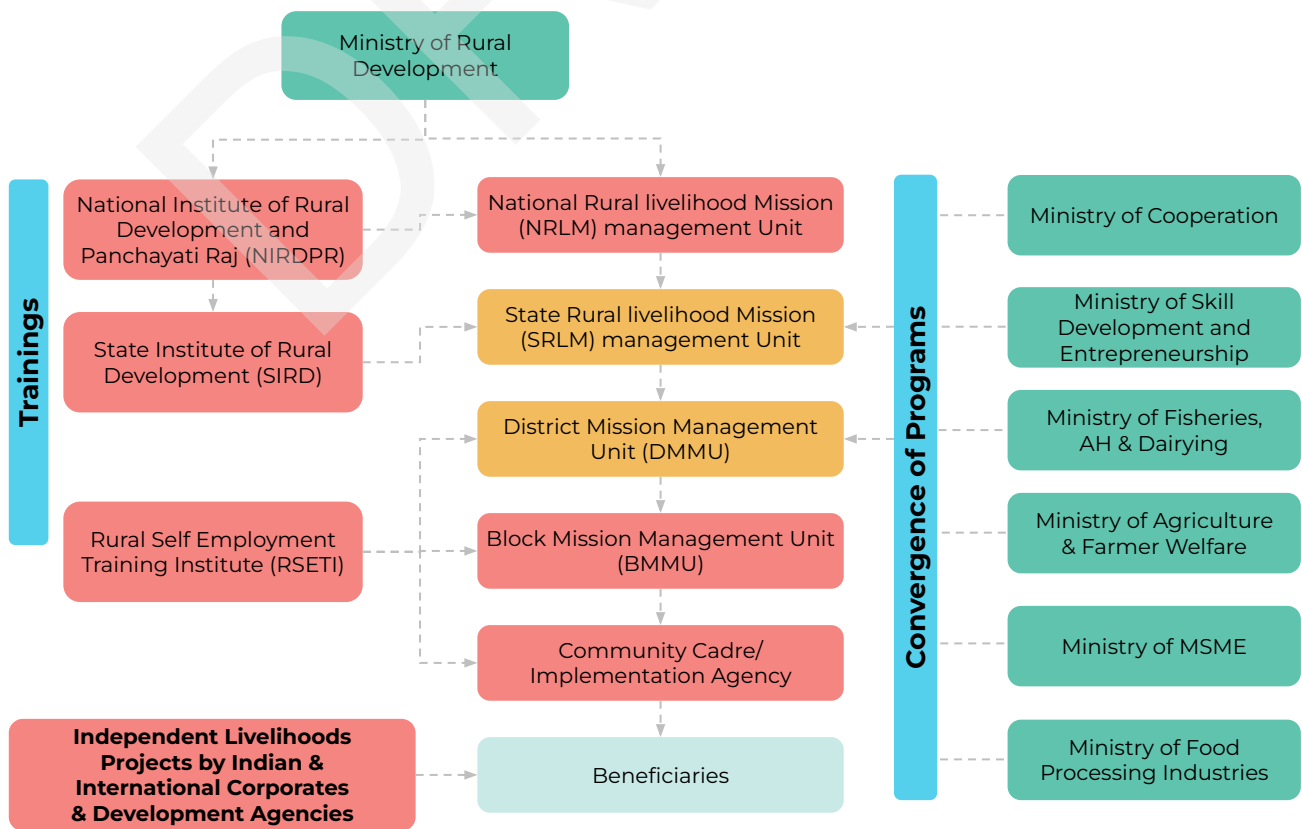
- State Rural Livelihood Missions (SRLMS)
- National, State, District and Block Mission Management units
- District Project Coordination Units
- Block Project Implementation Units
- Respective Dept. Support Unit
- Training Institutes
- Cluster Based Business Organizations (CBBOs)
- Farmer Producer's Organisations (FPOs)
- Small Farmers' Agribusiness Consortium (SFAC)

BENEFICIARIES

- Farmers
- Small & Marginal Farmers
- Rural Poor Household
- Self Help Groups
- Schedule Cast and Schedule Tribe
- Unemployed youth
- Farm labour
- Farmers Producer Organizations

Major Government Institutions in Farm-based Livelihoods Development Ecosystem

Figure 5: Government Institutions in Farm-based Livelihoods Development Ecosystem



Sub-sectoral Trends

Agriculture based livelihood

India is equipped with the second-largest arable land resources in the world. It is home to 20 agri-climatic regions and all 15 major climates across the globe.⁶ The country also possesses 46 of the 60 soil types found in the world.⁷ The variety of agro-climatic conditions supports production of wide variety of agricultural crops which largely includes food crops, cash crops and horticultural crops.

Total Work Force



Total Workforce
263,142,470



62.8%
Male Workforce
165,505,271



37.1%
Female Workforce
97,637,199

Source: Agriculture Statistics at a Glance 2022

Understanding the Agriculture Sub-sector through Facts:



Staple grains

- Staple grains accounts about 75% of the total food crop production (2021-22).⁸
- Rice is the top exported agricultural commodity contributing a share of **21% to the total agricultural exports** (2021-22).⁹



Food crops

- Production of food crops have observed **CAGR of 2.8%** in last 5 years (2017-22)¹⁰



Cash crops

- The total cash crop production in India in 2021-22 was at 477 MMT which grew at a **CAGR of 6.6%** between 2017-22. Sugar is the top export commodity among cash crops as it contributes **11% share** to the total Indian agricultural exports.¹¹



Horticulture crops

- India ranks 1st in production of many high value horticultural crops such as mango, banana, coconut, papaya, ginger & okra. Fresh fruits and vegetables accounts **2% share** in the total agricultural exports from the country (2022-23). The production has increased at a **CAGR of 2.6%** in the last 5 years (2017-22).¹²



Food processing

- The Indian food processing industry accounts for **32%** of India's total food market. The industry is estimated to be worth **USD 121 billion**, which makes it one of the largest industries in India. It ranks **5th** in terms of production, consumption and exports. The food processing industry employs about **1.7 million** people.¹³

Over the years, the agriculture production in India has increased its output to achieve global ranks. The sector holds immense potential to augment the quality and quantity of output by leveraging modern agriculture practices and upscaling crop production tailored for high-value markets like exports and processing industries.

Major Domains

Horticulture, food-processing, cash crops, field crops and sericulture

⁶ India Brand Equity Foundation (2022) Agriculture and allied Industries

⁷ India Brand Equity Foundation (2022). Agriculture and allied industries Business Opportunities in India

⁸ India Brand Equity Foundation (2022) Agriculture India

⁹ Department of Agriculture & Farmers Welfare, MoA & FW, Government of India, India (2022). Agricultural Statistics at a Glance 2022

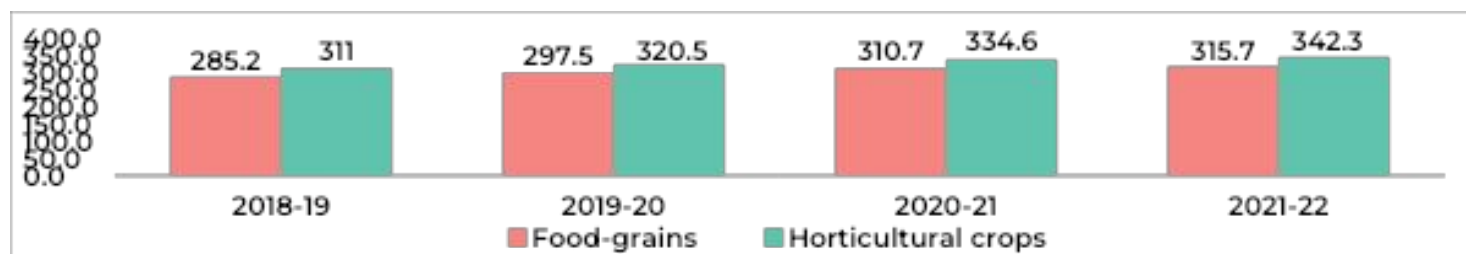
¹⁰ India Brand Equity Foundation (2022). Agriculture and allied industries Business Opportunities in India

¹¹ India Brand Equity Foundation (2022). Agriculture and allied industries Business Opportunities in India

¹² India Brand Equity Foundation (2022). Agriculture and allied industries Business Opportunities in India

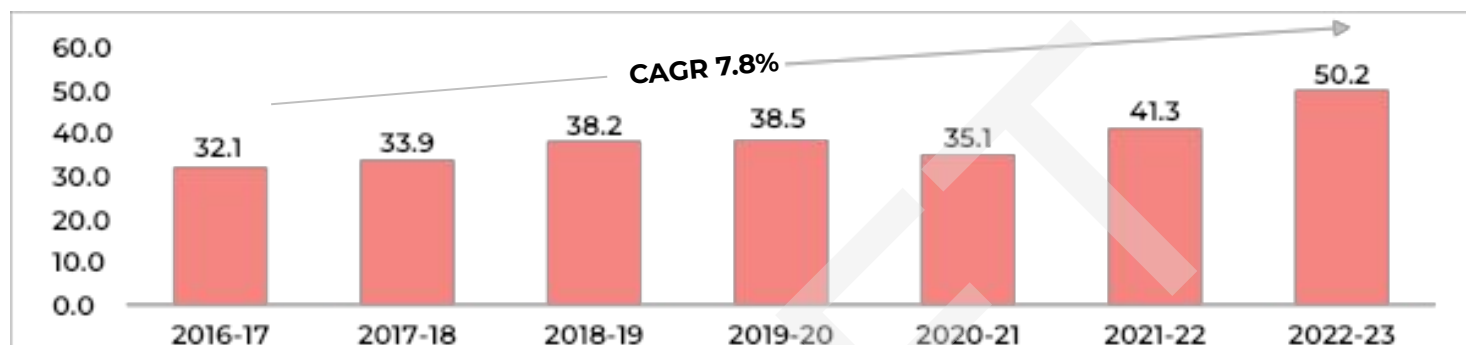
¹³ India Brand Equity Foundation (2022). Agriculture and allied industries Business Opportunities in India

Figure 6: Annual Agriculture Production Figures (in MMT)



Source: Agriculture Statistics at a Glance 2022

Figure 7: Agriculture Exports from India (in USD billion)

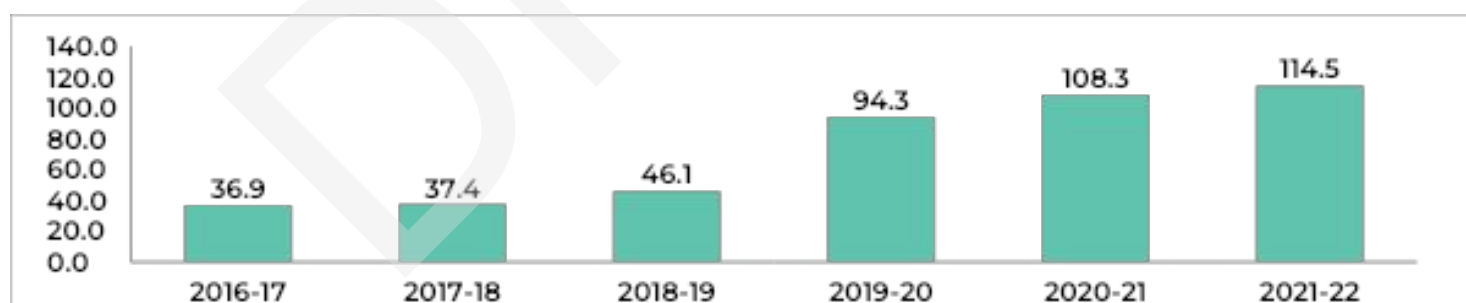


Source: Handbook of Statistics on the Indian Economy, 2022-23

Budget Trends

As per the budget 2023-24, INR. 1.25 L crore has been allocated to department of agriculture which accounts for 2.8% of the total union budget.¹⁴ The new budget provides higher impetus on regenerative agriculture (natural farming), inclusive growth, improved access to agriculture credit and inputs, digitization and technological development. The increase in expenditure over the past 4 years is on account of increase in the allocation for schemes such as modified interest subvention scheme (5%) and the PM-Fasal Bima Yojana (10%).

Figure 8: Actual Budget Expenditure by Ministry of Agriculture & Farmers Welfare (MoA & FW) (in '000 Cr)*



Source: Agriculture Statistics at a Glance 2022

*Budget allocation towards Department of Agricultural Research and Education (DARE) have not been included.

*05.02.2019 a new Ministry namely, Ministry of Fisheries, Animal Husbandry & Dairying had been created by carving out from the erstwhile D/o Animal Husbandry, Dairying and Fisheries under M/o Agriculture & Farmers Welfare.

¹⁴ Press Information Bureau News (2023) : Ministry of Agriculture & Farmers Welfare, Many key provisions for Agriculture and Farmers Welfare in the budget

¹⁵ Department of Agriculture & Farmers Welfare, MoA & FW, Government of India, India (2022), Agriculture Statistics at a Glance 2022

Animal Husbandry and Dairying based Livelihood

India occupies a significant position in livestock production sector across the globe. The country has a rich heritage of traditional livestock farming practice, which majorly includes rearing of cattle, buffalo, sheep, goat, pigs and poultry. The livestock sector provides nutrition, livelihoods and income opportunities to more than two-third of rural community.¹⁶ As per the 70th round of NSSO (2014), livestock farming is a principal source of income to 3.7 per cent of the agricultural households.

Total Work Force



Total Workforce
~80,000,000



-31%
Male Workforce
~24,800,000



-69%
Female Workforce
~55,200,000

Source: NLP 2013 and PIBNews, Cabinet. (2023)

*Rough estimates of workforce in dairy production through Cattle, Buffalo, Goat and Sheep rearing

Understanding the Animal Husbandry Sub-sector through Facts:



Production

- Livestock sector observed a **CAGR of 6%** in between 2016 to 2021.¹⁷
- The Livestock sector contributes **4.9% to the GDP** and 30.1% of agriculture GDP (2021).¹⁸



Workforce

- Animal husbandry (A.H.) sector supports livelihoods of more than **8 Crore Indian farmers**.¹⁹
- The A.H. and dairy sector provide around **50% direct & indirect employment to women** in the country which is the highest for any sector in the economy.²⁰



Dairy

- India is the **largest producer of milk** in the world, contributing **24% of global milk production** (2021-22).²¹
- Milk production grew at an **annual growth rate of 6.4%** over the past 8 years (2014 to 2022) while world milk production grew at 1.2% per annum.²²



Meat

- India ranks 8th in the world in terms of total meat production.²³
- The meat production from poultry is 4.5 million tonnes, contributing about 50.8% of total meat production.²⁴



Egg

- India **ranks 3rd** in the world in terms of total egg production.²⁵
- Total egg production in the country is 122 billion numbers.²⁶

¹⁶ [Livestock extension services \(2021\): Time to think beyond treatment and breed improvement. MANAGE](#)

¹⁷ [World Bank \(2022\): National Dairy Support Project \(NDSP\)](#)

¹⁸ [FAOSTAT Data \(2021\)](#)

¹⁹ [Ministry of Finance, Government of India \(2023\). Economic survey 2022-23 India Budget](#)

²⁰ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

²¹ [FAOSTAT Data \(2021\)](#)

²² [InvestIndia – Animal Husbandry and Dairying](#)

²³ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

²⁴ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

²⁵ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

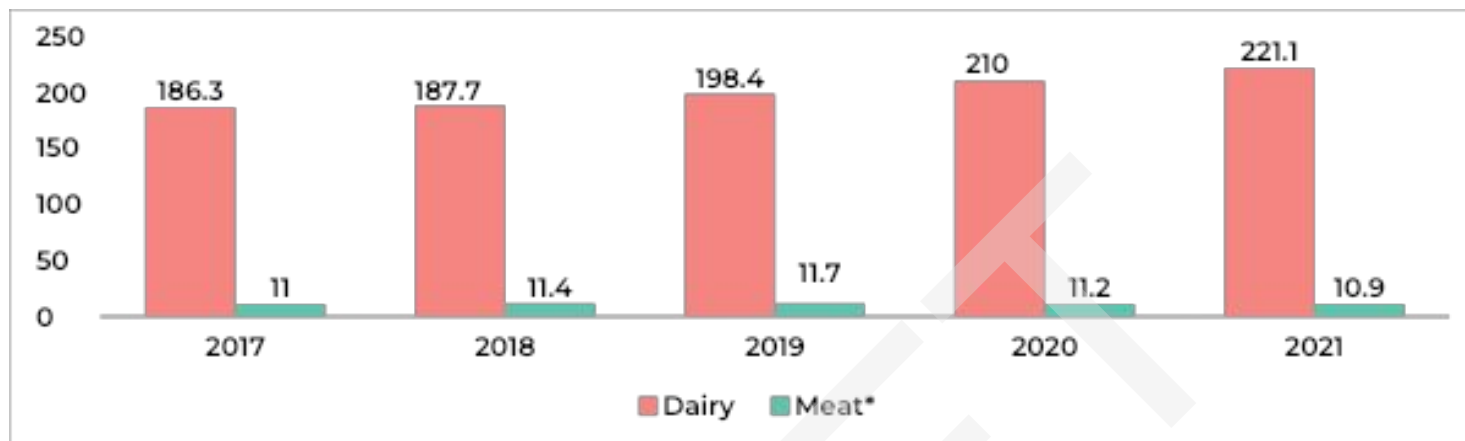
²⁶ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

- There were **5.4 million women** members in dairy co-operatives during 2020-21, accounting for 31% of total members.²⁷

- Total meat production in the country is 8.8 million tonnes, which is 2.3% increase from 2020 to 2021.²⁸

- Egg production in the country observed 6.7% increase from 2020 to 2021 and a **CAGR of 8%** during the period between 2016-2021.²⁹

Figure 9: Annual Dairy and Meat Production (in million tonnes)



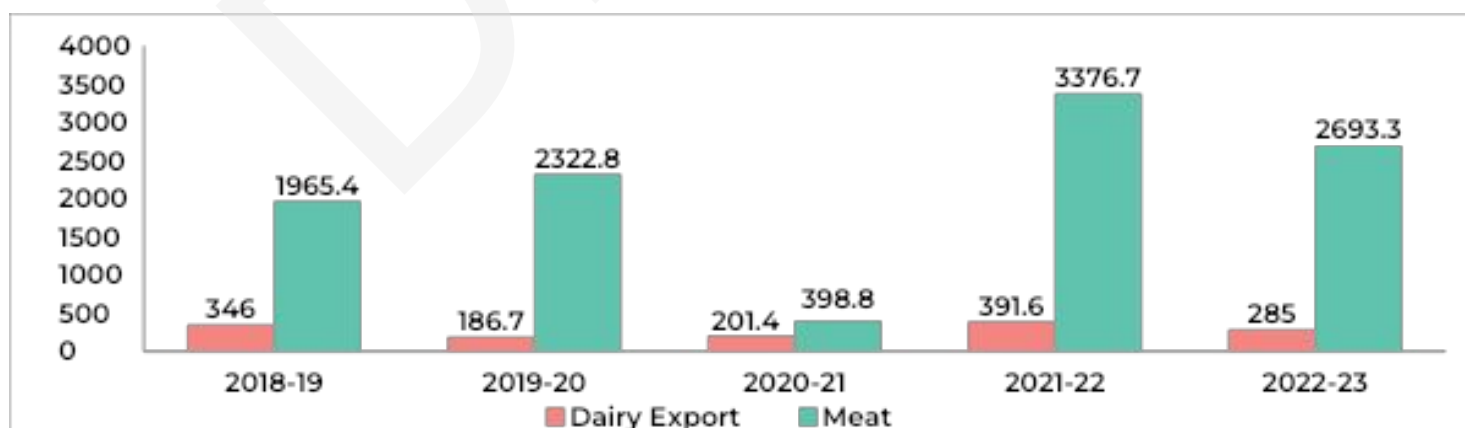
Source: [FAO \(2021\): NDDB](#). *Meat includes bovine, poultry, sheep/mutton, goat, pig meat, and wild game

Despite global standing in livestock production, the growth of Indian Animal husbandry sector is hindered by low-animal productivity, shortage of feed, disease outbreaks, inefficient rearing practices, lack of credit and market access, post-harvest losses, among others. Special development frameworks addressing these challenges shall be pivotal in doubling the farmer income and adoption of livestock farming as main occupation for livelihoods development..

Major Domains

Dairy farming, Goat and Sheep farming, Poultry farming and Piggery

Figure 10: Indian Export of Major Livestock Products



*Meat export data includes both meat and edible meat offal

Source: [APEDA \(2023\)](#) & Ministry of Commerce & Industry, India

²⁷ [World Bank: National Dairy Support Project \(NDSP\)](#)

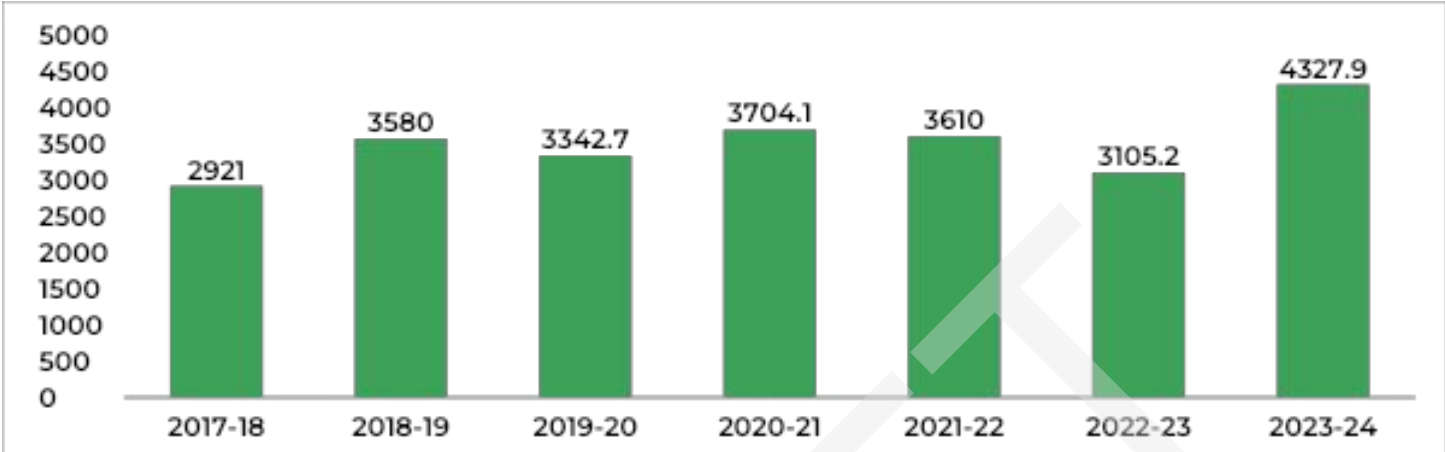
²⁸ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

²⁹ [Department of Animal Husbandry & Dairying, Government of India \(2022\). Annual Report 2021-22](#)

Budget Trends

Union budget 2023-24 has increased allocations to the Animal husbandry sector by 39.3% from the revised estimates of 2022-23. The new budget has also increased the agriculture loan target to INR. 20 L crores, with an emphasis on Animal husbandry sector.³⁰

Figure 11: Annual Budget Estimate for DAHD (INR Cr.)



Source: DAHD Annual Reports

Aquaculture based Livelihood

India is the world's second-largest aquaculture nation and the third-largest fish producer. More than 10% of the world's fish and shellfish species are found in India's rich and diverse fisheries, including deep seas, lakes, ponds and rivers.³¹

The country's extensive coastline, Exclusive Economic Zone (EEZ) and continental shelf region are home to marine fisheries resources. India has 2.3 million hectares of ponds and tanks, a coastline of 8,118 kms and brackish water area of 1.1 million which offers a large ground for fish farming and bolsters sustainable achievement of Indian blue economy goals. The sunrise sector is poised for sustainable growth through the blue economy mission as it also executes Sustainable Development Goal (SDG 14).³²

Total Work Force

The fisheries sector is a means of livelihood for ~28 million people in the country (2020-21)



Total Workforce
28,063,538



-55.8%
Male Workforce
15,665,630



-41.1%
Female Workforce
12,397,908

Source: Handbook on fisheries statistics, 2022

Understanding the Aquaculture Sub-sector through Facts:



Production



Export

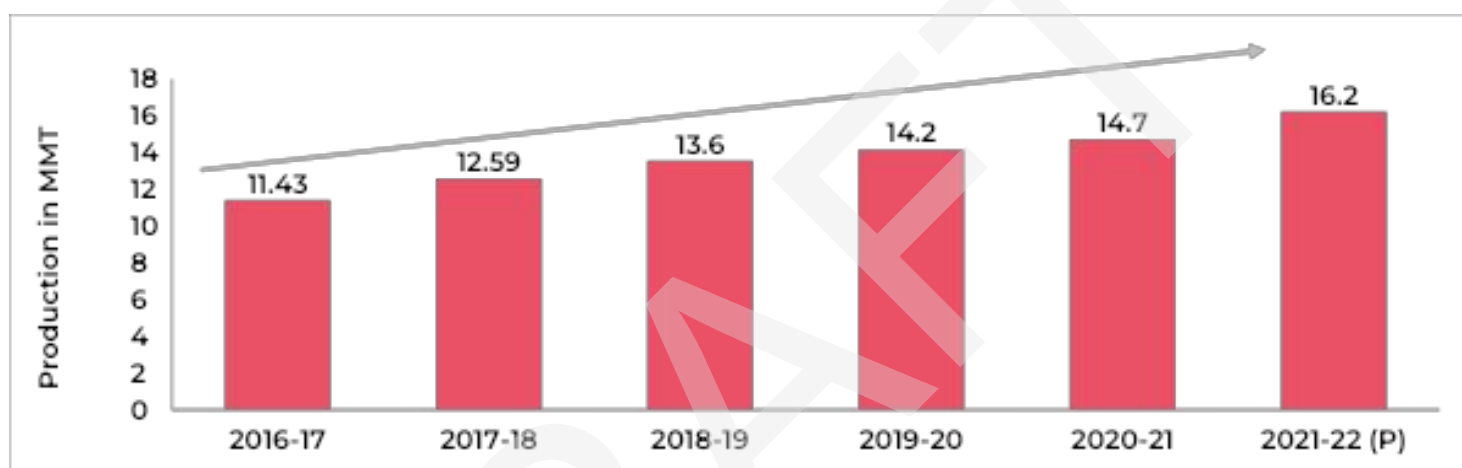


Workforce

- India accounts for around **8%** of the global fisheries production (2021-22).³³
- About **17%** of India's agricultural exports are fish and fish products. In 2021-22, the country exported marine products worth USD 7.7 billion, weighing 1.3 million MT. This marks an increase of 30% over the previous year (2020-21).³⁴
- India has 25,297 registered fishery cooperative societies which encompass nearly **3.8 million members**.³⁵

- Aquaculture sector observed a **CAGR of 7.3%** between (2016 to 2021).³⁶
- In 2021-22, India's total marine and inland fish production stood at 16.2 million metric tonnes (MMT), which includes 12.1 MMT and 4.1 MMT from the inland and marine sectors, respectively.³⁷
- India has the **3rd largest share** in the global aquaculture trade market.³⁸
- India mainly exports fish, frozen shrimps, squids, cuttlefish, dried items, chilled and live items. Among these, frozen shrimp is the largest exported marine product contributing to more than **53%** of the total quantity and **75.1% of the total export value**.³⁹
- In 2021-22, frozen fish, cuttlefish and squid contributed 6%, 3.6% and 4.9% of the total marine export value, respectively. This contributed 16.5%, 4.3% and 5.5% respectively to the total quantity exported in the same year.⁴⁰
- Marine exports from India are expected to reach **USD 14 billion** by 2025.⁴¹

Figure 12: Fish Production in India (2016-2021)



Source: [Handbook Fisheries Statistics 2023](#)

Of late, India's fisheries sector has diversified from traditional fishing to high-value aquaculture, including shrimp farming, mariculture and seaweed cultivation. However, low momentum in organized farming, digital integration, capital investments, and infrastructure are posing slowdowns to its growth. Nevertheless, targeted efforts to enhance the efficiency of the value chain have the potential to greatly improve the livelihoods of fish farmers.

Major Domains

Fisheries (Inland and Marine, Shrimp farming, Mariculture, Seaweed cultivation)

³⁰ Ministry of Fisheries, Animal Husbandry and Dairying, (2023-2024). Notes on Demands for Grants

³¹ [India Brand Equity Foundation Blog: The fisheries aquaculture sector of India](#)

³² [InvestIndia fisheries aquaculture](#)

³³ [Press Information Bureau \(2023\): Ministry of Fisheries, Animal Husbandry & Dairying](#)

³⁴ [Economic survey 2022-23: Statistical appendix](#)

³⁵ [Press Information Bureau: Cabinet news \(2023\)](#)

³⁶ [India Brand Equity Foundation marine products industry in India](#)

³⁷ [Department of Fisheries, Government of India \(2023\). Handbook on Fisheries Statistics 2022](#)

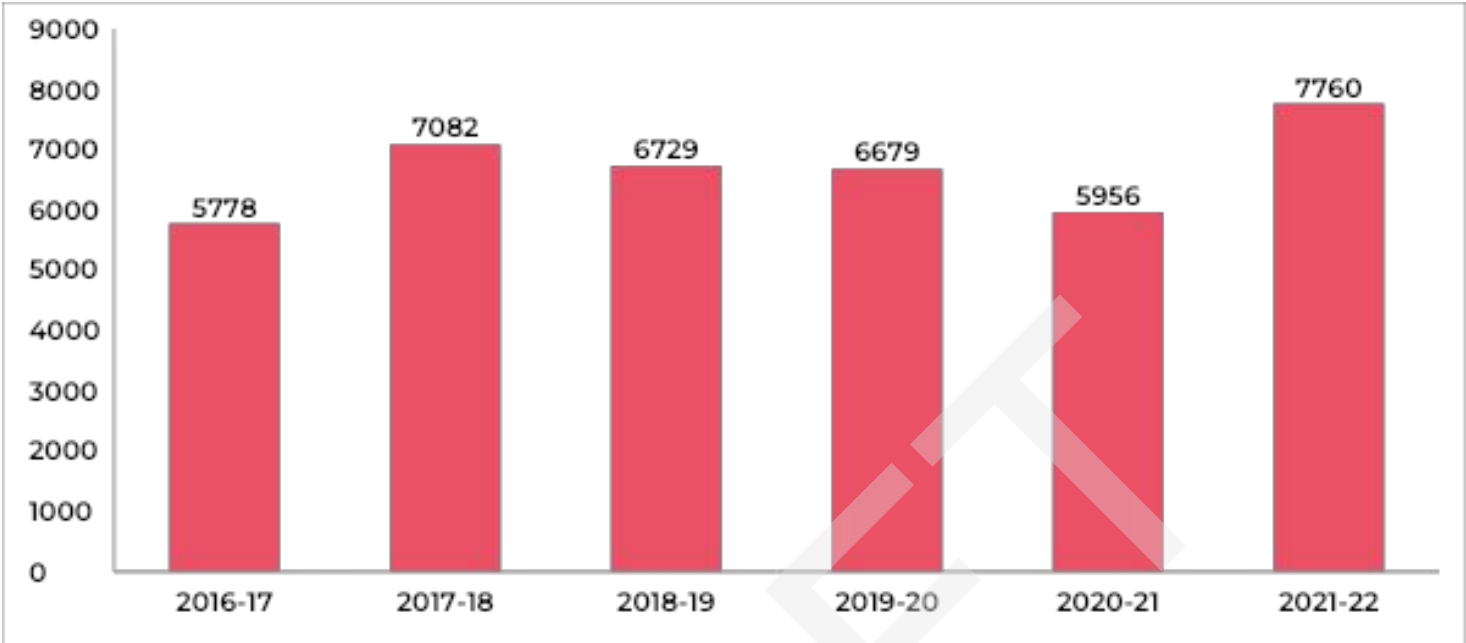
³⁸ [Project information memorandum: National Fisheries Development Board](#)

³⁹ [India Brand Equity Foundation marine products industry in India](#)

⁴⁰ [India Brand Equity Foundation marine products industry in India](#)

⁴¹ [India Brand Equity Foundation marine products industry in India](#)

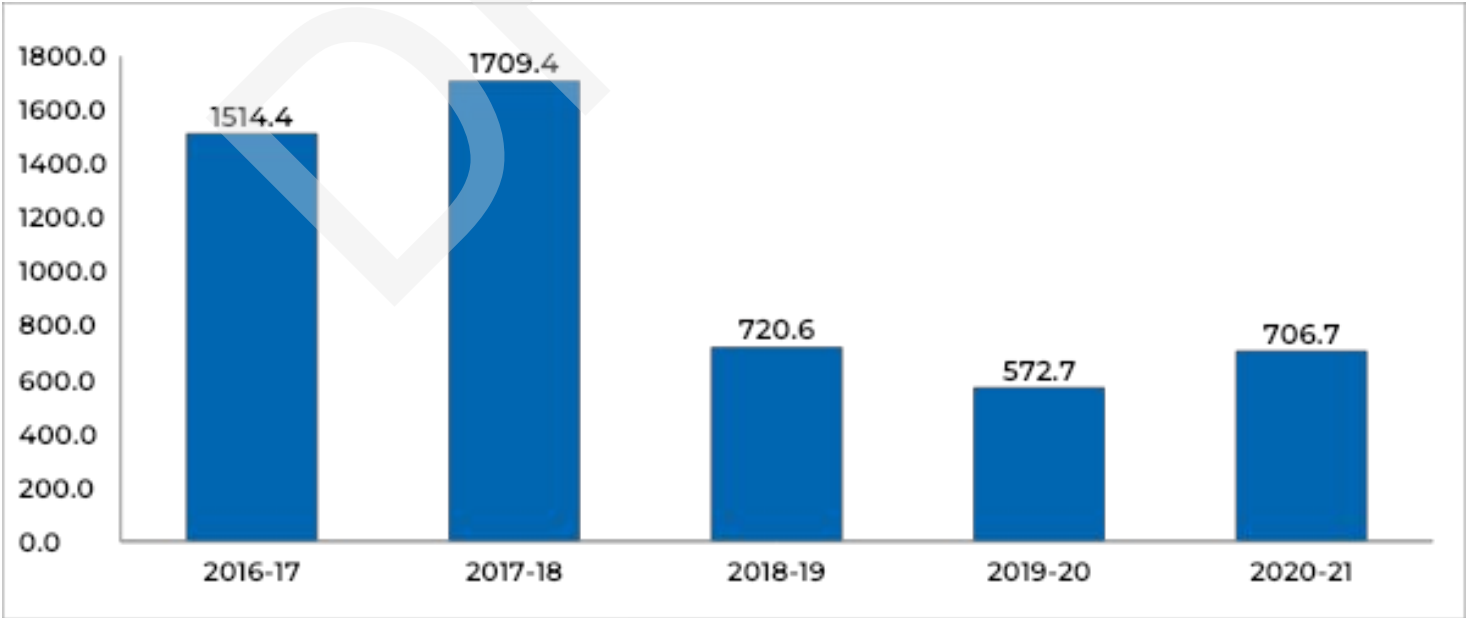
Figure 13: Export trends of marine products from India (2016-2021)



Source: [MPEDA annual report \(2021-22\)](#)

The department of fisheries has been allocated INR 2248,7 Cr for 23-24, which marks an overall increase of 38.5% from 22-23 budget allocation of INR 1624.8 Cr.⁴²

Figure 14: Annual Fund Released for Development of Fisheries Sector (in INR Cr)



Source: [Handbook Fisheries Statistics 2023](#)

⁴² Press Information Bureau News (2023) : Ministry of Fisheries, Animal Husbandry & Dairying

Opportunities and Challenges



AREAS	CHALLENGES ⁴³	OPPORTUNITIES
Small and fragmented holdings ⁴⁴	Small land holding limits production, diversification, effective resource utilisation, bargaining power and building scale for linking with forward markets. 85% of the Indian farmers have an average land-holding of 1.08 hectare. ⁴⁵	• Production of high value specialty produce (crops/aquaculture/livestock produce) for niche markets. • Promotion of integrated farming (e.g. duck, poultry and fish farming), mixed-farming and intensive farming methods (e.g. Recirculation aquaculture systems) to maximise utilization of available resources. • Collectivisation of produce to build scale and improving bargaining power • Training farmers to produce reconstituted feed items from primary crop residue and plantation of dual-purpose forage crops
Resource mobilization ⁴⁶	• Unequitable utilization of available resources at scale • Low women participation • Lack of formal income generation activities	• To supplement income generation through agro tourism (est. global agro-tourism CAGR of 13.4% between 2020-2027) ⁴⁷ • To promote gender inclusivity across key value chain activities such as decision making, financial upkeep and others. • Formation of village level societies/co-operatives for collective production.
Social capital and limited collectivization ⁴⁸	• Low bargaining power of individual farmer and scale in production • Low comprehensive handholding provision for farmers • Limited collectivization. Only 17 lakh farmers registered as Farmer Producer Organization (FPO) members. ⁴⁹	• Strengthening collectivisation of producers in the form of producer groups (PGs), Self-help groups (SHGs) and FPOs/FPCs • Formation of apex body for FPOs fostering an inclusive representation of women and marginalised groups as key human capital Capacity building of Krishi/Padhu Sakhis along with follow up of their work shall be ensured for sustainability
Access to finance	• Producers have poor access to credit facilities. Only 60% farmers have access to institutional credit. ⁵⁰ • Inability to access credit due to inadequate financial literacy and collateral	• Develop innovative credit infusion methods/investment schemes. • Promoting financial literacy, development of producer-friendly loan products and streamlining the process of availing credit support and its disbursement
Lack of infrastructure	• Post-harvest losses • Low processing of farm produce at farm-gate. Only 2% processing in F&V, 35% in milk, 21% in meat and 8% in marine products. ⁵¹	• Private sector participation in storage, warehousing and value addition related infrastructures • Investment in processing facilities near farm gate for collective primary and secondary processing and packaging. • Development of digital public infrastructure for traceability

⁴³ World Bank. (2012) India: Issues and Priorities for Agriculture Central Institute for Research on Goats Makdhoom (2018) : Doubling rural farmers' income through goat farming in India: prospects and potential

⁴⁴ World Bank. (2012) India: Issues and Priorities for Agriculture

⁴⁵ Press Information Bureau News (2019): Ministry of Agriculture & Farmers Welfare – Categorisation of farmers

⁴⁶ GiZ (2022). Not reinventing the wheel. Compilation of successful livelihood models from rural India.

⁴⁷ MANAGE, Hyderabad (2023) Agritourism: The Stakeholders-Focused Journey in Maharashtra.

⁴⁸ World Bank group (2019): Labor and Welfare Impacts of a Large-Scale Livelihoods Program: Quasi-Experimental Evidence from India

⁴⁹ The Indian Journal of Agricultural Sciences (2022). Progression of Farmer Producer Organisations in India

⁵⁰ NABARD (2022): Trends and Patterns in Agriculture Credit in India: A District Level Analysis of Uttar Pradesh.

⁵¹ Ministry of Food Processing Industries (n.d.): food availability map of India

Technology interventions and adoption ⁵²	Poor affordability and access to new technologies. The level of farm mechanization in India is nearly at 40%.⁵³	- Development of one-stop platform to provide extension support on best practices (Package of Practices), market information, crop advisory and others - Encouraging producers to adopt new technologies via extension support on its use and benefits. - Intensifying technology use for value chain development and support in acceleration and adoption of innovation - Fostering successful adaptation of Custom Hiring Center (CHC) facilities via extension and initial handholding support
Information asymmetry	Low access to information on market dynamics, efficient production and management practices, support schemes/policies etc. There are only 1,361 APMCs registered with e-NAM. ⁵⁴	- To establish a technologically robust platform for providing market information (commodity prices, demand-supply trends, linkage advisory etc), extension support and more. to the producers - Extend the reach of field level guidance and implementation support agencies - Support in acquiring necessary certifications (e.g. FSSAI)
Lack of diversification ⁵⁵	Reliance on traditional farm-based products and low focus on high value markets.	- Promoting diversification into new high value crops, livestock, and fisheries through capacity building initiatives can potentially increase the net return by 2-5 times.⁵⁶ - Incentivizing expansion to high value markets
Market linkages ⁵⁷	- No. of intermediaries in the value chain reduces the farmer's profit while escalating the end prices. - Lack of quality standardization	- Maximisation of farmers share in consumer price through value chain-based approach - Promotion of branding and marketing activities shall foster widespread recognition and linkages with new market channels such as export markets. Share of Indian farm export and import in global trade in 2021 were 2.4% and 1.7%, respectively.⁵⁸
Natural calamities	Farm based livelihoods are vulnerable to climatic actions, livestock disease outbreak and other risks. Climate change is predicted to reduce farm yield such as wheat, rice and maize yield by 19.3%, 20% and 18% by 2050. ⁵⁹	Development of resilience towards adversities through affordable insurance products, promotion of sustainable production practices and climate resilient inputs, stringent biosecurity, disease surveillance, immunization measures and other risk mitigation strategies.
Migration	Increased migration of youth from rural to urban areas. Nearly 25-30 people migrate from rural to urban areas every minute. ⁶⁰	Promoting entrepreneurship, initial handholding, training and capacity building and business incubation support in rural areas.
Convergence ⁶¹	Beneficiaries are unable to encash optimum benefits from existing schemes due to lack of awareness on available support schemes and complex regulatory environment	- Integration of policy programs and schemes for achieving synergistic outcomes and expanding the reach of benefits. (e.g. NREGA in Maharashtra converged with NHM and NHB schemes) - Introduction of single window system for all related schemes by different departments. - Inviting participation of corporate entities to act in synergy with existing public schemes for extending support services for livelihoods improvement.

Table 1: Opportunities and challenges of farm-based livelihoods sector

⁵² World Economic Forum. (2022) Insight report: Using Technology to Improve a billion Livelihoods

⁵³ IIM Ahmedabad. (2023). Mechanization in Agriculture: Assessment of skill development gap and adoption of labour-saving technologies

⁵⁴ e-NAM: APMCs

⁵⁵ GiZ (2022). Expanding Livestock based livelihood opportunities through MGNREGA: An Exploration

⁵⁶ Indian Journal of Agricultural Sciences (2018). Integrated farming system in India: Current status, scope and future prospects in changing agricultural scenario.

⁵⁷ GiZ (2022). Not reinventing the wheel. Compilation of successful livelihood models from rural India

⁵⁸ Ministry of Agriculture & Farmers Welfare (n.d.): Overview of Agriculture Trade Policy. Promotion and Logistics Development Division

⁵⁹ Press Information Bureau News 2023): Ministry of Agriculture & Farmers Welfare- Impact of Climate Change on Agriculture

⁶⁰ InvestIndia Blogs (2018): India preparing biggest human migration planet

⁶¹ GiZ (2022). Not reinventing the wheel. Compilation of successful livelihood models from rural India

Cross-cutting Themes

Figure 15: Cross-cutting themes



Government Interventions



Key Policies, Themes/Acts/ Programs	Ministries/ Govt. Agencies	Target Group	Key Features
Rashtriya Krishi Vikas Yojana -(RKVY) Year of Initiation - 2007	Ministry of Agriculture and Farmers' Welfare	Rural Youth, Farmers, Students, Micro-enterprises	It aims to make farming a remunerative economic activity by strengthening the farmers efforts, risk mitigation and promoting value addition and agri-business entrepreneurship. The scheme also empower youth through skill development, innovation and Agri entrepreneurship-based agribusiness models.
Deendayal Antyodaya Yojana-National Rural Livelihood Mission (DAY-NRLM) Year of Initiation - 2011	Ministry of Rural Development	Rural Populace under poverty (esp. women)	Mahila Kisan Sashaktikaran Pariyojana (MKSP) Sub-Scheme empowers women in agriculture by strengthening community institutions of poor women farmers, supporting institutionalization of SHGs and development of financial and governance linkages for robust institutionalization among them. It provides intensive capacity building and training of SHG members for collectivisation, primary & secondary processing of farm produce and creation of direct market linkages. It also provides funds to strengthen institutional and financial management capacity and build their track record to attract mainstream bank finance.
National Programme for Dairy Development (NPDD) Year of Initiation - 2014	Department of Animal Husbandry and Dairying	Dairy farmers, Dairy co-operative and collectives	The scheme aims to create and strengthen infrastructure for production of quality milk including cold chain infrastructure, milk procurement, processing and marketing, training of dairy farmers, strengthening of dairy cooperative societies/Producers Companies at village level, providing technical input services and more.
National Livestock Mission (NLM) Year of Initiation - 2014	Ministry of Fisheries, Animal Husbandry and Dairying	Dairy producers and Dairy development institutions	The Sub-Mission on extension and Innovation aims to incentivise Institutes, Universities, Organizations carrying out research and development related to extension activities, livestock insurance and other innovations.

Mission for Integrated Development of Horticulture (MIDH) Year of Initiation - 2014	Ministry of Agriculture and Farmers Welfare	Farmers and Farmer collectives, aspiring farmers and unemployed youth	The scheme aims to develop capacity by encouraging farmer aggregation, improving productivity through new technologies and support skill development for rural youth in horticulture and post-harvest management.
Entrepreneurship Development & Employment Generation (EDEG) -component of National Livestock Mission Year of Initiation - 2014	Ministry of Agriculture and Farmers' Welfare	Farmers, SHGs, Joint Liability Group (JLGs), entrepreneurs, NGOs, Companies, Cooperatives, FPOs	The scheme supports employment generation through entrepreneurship development in livestock sector and capacity building of livestock owners through quality extension service.
Pradhan Mantri Kisan SAMPADA Yojana (PMKSY) Year of Initiation - 2015	Ministry of Food Processing Industries	FPOs, Entrepreneurs, Private Firms	The scheme covers components for development of Integrated Cold Chain and Value Addition Infrastructure, Infrastructure for Agro-Processing Clusters, Creation of Backward and Forward Linkages, Creation/Expansion of Food Processing & Preservation Capacities and Food Safety and Quality Assurance Infrastructure.
ASPIRE (Promotion of Innovation, Rural Industries & Entrepreneurship) Year of Initiation - 2015	Ministry of Micro, Small & Medium Enterprises	Farmer and farmer collectives, MSMEs	The scheme imparts skill development & incubation support through Livelihoods Business Incubator (LBI) for promoting entrepreneurship and employment generation in agro-rural sector with special focus on rural/underserved areas.
Shyam Prasad Mukherji Rurban Mission Year of Initiation- 2016	Ministry of Rural Development	Rural population	The scheme is aimed at developing rural areas by provisioning of economic, social and physical infrastructure facilities.
Dairy Processing and Infrastructure Development Fund (DIDF) Year of Initiation - 2017	Ministry of Fisheries, Animal Husbandry and Dairying	Dairy Farmers, collectives and Dairy organizations	The scheme provides funds for Dairy development and technological advancements in dairy infrastructure including milk cold chain, marketing infrastructure, Renewable energy infrastructure, among others.
Agriculture export policy Year of Initiation - 2018	Ministry of Agriculture and Farmers' Welfare	Farmer collectives, Agribusinesses	The scheme provides an institutional mechanism for pursuing market access, tackling barriers, enable farmers get benefit of export opportunities and deal with Sanitary and phytosanitary (SPS) issues.

PM Kisan Samman Nidhi Year of Initiation - 2019	Ministry of Agriculture and Farmers' Welfare	Land holding farmer families	The scheme supplements the financial needs of the SMFs in procuring various inputs to ensure proper crop health, appropriate yields and farm income.
PM Matsya Sampada Yojana Year of Initiation - 2020	Ministry of Fisheries, Animal Husbandry and Dairying	Fish farmer, collectives (SHGs, JLGs, FPOs), Entrepreneurs, Private Firms	The scheme provides financial assistance to develop fishing infrastructure including cold chains, fish production and processing units, packaging facilities among others. The scheme aims to modernise and strengthen the value chain, improve production quality and build a robust fisheries management and regulatory framework.
Pradhan Mantri Formalisation Of Micro Food Processing Enterprises Scheme Year of Initiation - 2020	Ministry of Food Processing Industries (MoFPI)	FPOs, SHGs, cooperatives, any Government agency, or private enterprises	The common infrastructure scheme support development of common infrastructure for sorting, grading, assaying, storage common processing, packaging and marketing at the farm-gate. It also supports training the members of SHGs/FPOs/Cooperatives related to micro-food processing enterprises in processing activities, provides seed capital to SHG for working capital requirements.
Animal Husbandry Infrastructure Development Fund (AHIDF) Year of Initiation - 2020	Ministry of Fisheries, Animal Husbandry and Dairying	Individual entrepreneurs, private companies, MSME, Farmers Producers Organizations (FPOs), companies	The scheme incentivises investments to establish value addition infrastructure in Livestock production sector, breed improvement technology, waste management and more.
National Agriculture Infra Financing Facility Year of Initiation - 2020 man Nidhi	Department of Agriculture & Farmers Welfare	Farmers (including FPOs), PACS (Primary Agri. Cooperative Societies), Agri entrepreneurs	To mobilise a medium - long term debt financing facility for investment in viable projects relating to postharvest management Infrastructure and community farming assets through incentives and financial support.

Table 2: Major Government Interventions in Farm-based Livelihoods Sector



Farm level diversification to enhance livelihoods:

- 85% of the farmers are small and marginal farmers and they possess only **44%** of the total operational land.⁶² Limited land ownership demands to engender greater economic productivity per unit area of land. Capacity building and training of producers should go beyond traditional production practices and diversify production as per the emerging trends, consumer preferences, and potential niche markets. Information on economic benefits of diversification and value addition for enhanced income and risk resilience shall be disseminated.
- Fostering transition from subsistence farming to commercially oriented farming by switching from production of low-value products to high value farm products.
- Technical and financial support for implementing integrated production systems (e.g. Encourage integrated farming systems with integration of livestock, poultry, fish farming among others) or mixed farming systems to optimise resource utilization and diversify income streams.



Promoting access to credit and insurance facilities:

- Despite significant government subsidy programs, nearly **95 million farmer** households are not covered under any insurance product.⁶³ Insurance penetration stands at nearly **22%** of the total number of farmers covered and **30%** of the gross cropped area.⁶⁴ Introduction of affordable insurance products shall mobilise the producers to avail low-cost insurance and credit facilities. Innovation in credit outreach and risk management products to spread the risk across the system (group lending vs individual lending).
- Improved access to affordable credit facilities by expanding the reach of interest subvention schemes, credit guarantee funds, farmer-friendly loan products etc. through rigorous extension support.
- Promote capacity building and training of producers on financial literacy for seeking benefits of existing support schemes.
- Encourage organization of farmers into collectives (e.g. FPOs, cooperatives and others) for availing multiple policy-level benefits while extending those benefits to all member producers.



Promoting last mile adoption of digital technologies:

- Digital penetration in agriculture is estimated to grow at a **CAGR of 22.5%**.⁶⁵ End user centric digital solutions to bridge information gaps (weather forecast, market information, best production practices and more), improve access to support services, enhancing market linkages, improving access to financial services and empowering farmers with real-time information and knowledge shall be promoted.
- Incorporating farm-level traceability systems to minimize losses, ensure product quality, consumer demand and higher returns.
- Establishing direct linkages with stakeholders. E.g., through e-marketplaces that connect producers directly with buyers (including retailers, distributors, and consumers).
- Educating producers about the usage and benefits of digital systems and emphasizing their convenience, security, and potential for enhancing livelihoods opportunities.

⁶² Indian Journal of Agricultural Sciences (2018): Integrated farming system in India - Current status, scope and future prospects in changing agricultural scenario

⁶³ World Bank (2012): Improving Farmers' Access to Agricultural Insurance in India Policy Research Working Papers

⁶⁴ TCS White paper (2021): Crop Insurance: Improving Business Value Using Technology Interventions

⁶⁵ Strategic Investment Research Unit (SIRU). InvestIndia (2022): Digitalisation of Agriculture in India :

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Fostering common and shared infrastructure/machinery facilities

- Development of affordable custom hiring centres (CHC) or shared infrastructure/farm-machinery and ensuring its optimal utilisation to address low mechanization among small and marginal farmers. The farm-machinery sector has observed a growth rate of **3.6%** (2008-2018) and it is forecasted to grow at a **CAGR of 6%** by 2025.⁶⁶
- There are nearly 24,000 FPOs/FPCs registered in the country.⁶⁷ Fostering producers' aggregation/collectivization models to build scale. Collectivisation shall improve producers' bargaining power, facilitate technology adoption and access to required inputs/infrastructure.
- Exploring public-private partnerships with farm-machinery and equipment providers to offer CHC services to small and marginal farmers at affordable rates.(for e.g. EM3 Agri-services ties up with Bank of Baroda to provide affordable credit for farm machinery services)



Focusing on convergence with other programmes

- Identifying and prioritizing key sectors having direct or indirect impact on farm-based livelihoods (such as rural infrastructure, skill development, market linkages etc.), and fostering collaboration and coordination among different government departments, agencies, and ministries responsible for implementing programs in identified sectors. Inter-sectoral approaches shall optimise efforts and enable better planning and effective investments for livelihood development. Convergence of (National Rural Employment Guarantee Act) NREGA with Swarnjayanti Gram Swarozgar Yojana (SGSY), District Rural Development Agency (DRDA) and social forestry division of Assam is one such example.⁶⁸
- Aligning farm-based livelihoods development programs with other relevant sectors to create durable assets, maximise their impact, achieve synergistic outcomes and address interrelated challenges.
- To ensure regular review and update to policy programmes based on feedback from stakeholders and evolving challenges.



Value chain development approach

- Design and development of business models to foster a win-win situation for multiple stakeholders in the value chain. (e.g. convergence in agri-input shall ensure low cost, consistent supply, advice on the use of inputs which optimise production quality and overall profitability).
- Public Private Partnership (PPP) model of development to establish modern agri-business infrastructure such as mega-food parks, agri-zones and agri-tourism (e.g. Organic farm and farm tourism center, Jorhat)⁷⁰
- Fruits and vegetables in India incur post-harvest losses ranging from 5 to 16% . Development of post-harvest supply chain infrastructure (warehousing and logistics) to address the shortfall in storage and distribution facilities and avoid spoilage of perishable farm produce.⁶⁹ Cold chain infrastructure is expected to grow at a **CAGR of 14.3%** (2022-2027) in the country.⁷¹
- Encouraging different models of farming e.g. contract farming, under which there is an agreement between a buyer and farmers, which establishes conditions for the production and marketing of a farm product or products.
- Handholding support to producers and collectives to establish linkages with high value forward markets such as food processors, exporters, institutional buyers and others.
- Ensuring effective utilisation and implementation of value chain development support schemes and infrastructure through widespread awareness programmes.

⁶⁶ Agriculture Skill Council of India (2020). Qualifications handbook

⁶⁷ National Association for Farmer Producer Organisations (2023) State of sector report – Farmer producer organizations in India 2023

⁶⁸ UNDP and Ministry of Rural Development – Department of Rural Development (2013). Report on Convergence Initiatives in India – An Overview

⁶⁹ Agriculture Skill Council of India (2020). Qualifications handbook

⁷⁰ MANAGE (2018). Good Practices in Agricultural Extension and Advisory Services with Agripreneurship

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